

Lab 1 - Requirements Table: Database Query Performance Profiler (Problem Statement - 44)

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Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse Postgres/MySQL EXPLAIN plans, highlight sequential table scans on large tables, and recommend optimal index column definitions.	High	Pass: Missing composite index identified on filtered columns. Fail: Query plan with sequential scan marked optimized.	Core value proposition of the tool; without accurate plan parsing no useful recommendation can be produced.
FR-002	Functional	The system shall ingest slow query logs from Postgres and MySQL sources, normalize them into a common schema, and timestamp each record on arrival.	High	Pass: A log entry from either database engine appears in the normalized log table within 5 seconds of ingestion with all required fields populated. Fail: Any required field (query text, duration, timestamp, database) is null or the entry is dropped.	Reliable ingestion is the entry point for every downstream feature; malformed or lost entries invalidate all later analysis.
FR-003	Functional	The system shall generate a weekly performance optimization digest summarizing the top 10 slowest queries, associated index recommendations, and estimated performance impact.	Medium	Pass: A digest is generated every Monday at 06:00 containing exactly the 10 highest-duration queries from the prior 7 days with a recommendation for each. Fail: Digest is missing, generated late by more than 30 minutes, or omits a recommendation for a flagged query.	Gives the Database Administrator and Backend Lead a periodic, low-effort summary instead of requiring manual log review.
FR-004	Functional	The system shall allow a Database Administrator to mark an index recommendation as "Applied", "Dismissed", or "Pending" and retain that status for future digests.	Medium	Pass: Marking a recommendation as Applied removes it from subsequent weekly digests and the status persists after a system restart. Fail: A dismissed or applied recommendation reappears in a later digest.	Prevents repeat noise for recommendations already actioned or intentionally rejected, keeping the digest relevant.

FR-005	Functional	The system shall send a real-time alert to the Database Administrator when a query's execution time exceeds a configurable critical threshold (default 5 seconds).	High	Pass: A query exceeding 5 seconds triggers an alert delivered within 60 seconds of ingestion. Fail: A qualifying query produces no alert, or an alert fires for a query under the threshold.	Slow-query digests are weekly, but severe regressions (e.g., a lock-inducing query in production) need immediate escalation, not a delayed report.
NFR-001	Nonfunctional	Slow query log ingestion must handle up to 5,000 log records per minute without dropping entries or slowing host DB.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	The profiler must not become a source of the very performance problems it is meant to diagnose.
NFR-002	Nonfunctional	The system shall retain 90 days of historical query log and digest data, and remain available for query submission and dashboard access at least 99.5% of the time during business hours (08:00-20:00 local).	Medium	Pass: A rolling 90-day dataset is queryable for trend comparisons, and uptime monitoring over any 30-day window shows availability $\geq 99.5\%$ during business hours. Fail: Historical data older than 90 days is unexpectedly missing, or measured uptime falls below 99.5%.	Trend analysis (e.g., "is this query getting slower over time") and DBA trust in the tool both depend on consistent history and availability.